

# 4.7.58

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**Question:** Find the distance between the parallel lines  $3x - 4y + 7 = 0$  and  $3x - 4y + 5 = 0$ .

**Solution:** Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

The given lines can be expressed in matrix form as

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -7 \quad \text{and} \quad \begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -5 \quad (2)$$

Here, the normal vector is

$$\mathbf{n} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \quad c_1 = -7, \quad c_2 = -5 \quad (3)$$

The distance between two parallel lines is

$$d = \frac{|c_1 - c_2|}{\|\mathbf{n}\|} \quad (4)$$

$$\|\mathbf{n}\| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5 \quad (5)$$

$$|c_1 - c_2| = |-7 - (-5)| = |-2| = 2 \quad (6)$$

Therefore,

$$d = \frac{2}{5} \quad (7)$$

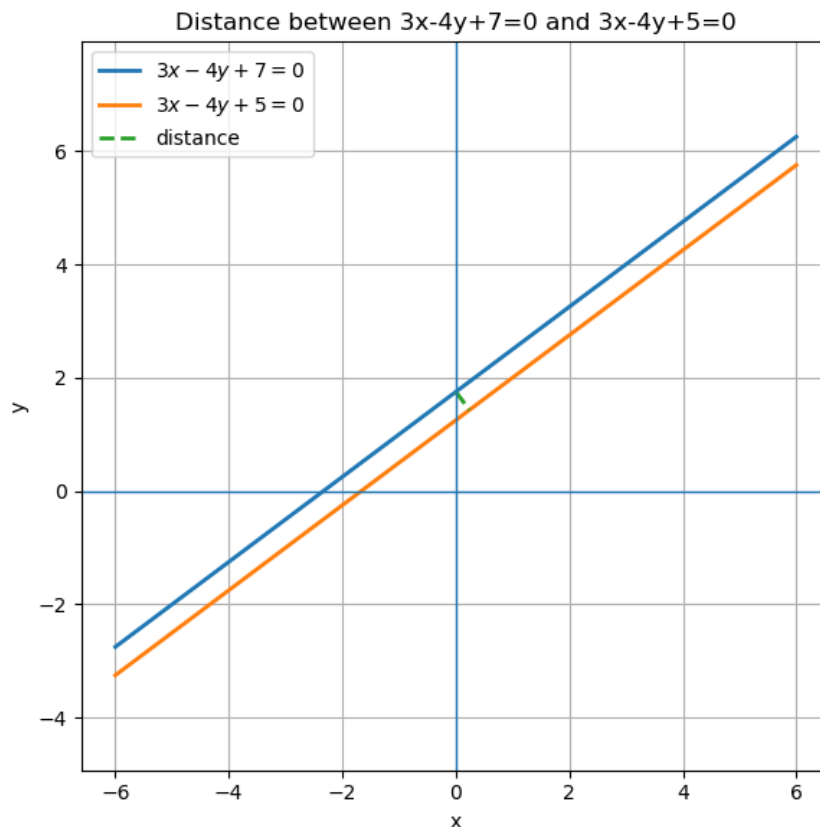


Fig. 0: Distance between two Parallel Lines